G H Raisoni International Skill Tech University, Pune.

Question Paper

Continuous Assessment Examination-I (CAE-I), Summer 2025

Programme: B Tech | **Year: First** | **Semester: II**
Max Marks: 20 | **Duration: 60 mins.**
Course Name: Digital Electronics | **Course Code: 24BTCL203**

Instructions:

1. You are permitted to use calculators during the exam.
2. Clearly label each question and ensure your answers are organized.
3. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q.No. 2C, etc.

Note: This question paper addresses below Course Outcomes (CO's):

CO1	To Demonstrate a clear understanding of binary number system, Boolean algebra, and the principles of digital logic
CO2	Ability to design, analyze, and optimize combinational and sequential logic circuits using logic gates and Boolean algebra.
CO3	Gain a thorough understanding of digital systems, including binary representation, Boolean algebra, and logic gates.

Q. No.	Sub Q. No.	Question	Marks
1		**Solve any TWO**	
	A	**I.** What is a digital signal? Explain how digital signals are different from analog signal. II.Write the truth table for an Exclusive-OR (XOR) gate	05
	B	I. Describe about universal gates. II. Derive basic gates using any of the universal gate.	05
	C	I. What is the main difference between one's complement and two's complement in terms of arithmetic operations? II. Find one's and two's complement of 1744.	05
	D	Do the conversion for: i. $(1011)_2 = (_____)_8$ ii. $(53)_{10} = (_____)_2$ iii. $(53)_{10} = (_____)_{16}$ iii. $(10010)_2 = (_____)_{10}$ v. $(AC)_{16} = (_____)_{10}$	05
2		**Solve any TWO**	
	A	i. Explain about K-Map? ii. Write the rules to solve and represent the K-Map.	05
	B	i. Solve the Boolean function using K-map F(A, B, C) = ∑(1, 3, 6, 7). ii. For above function realize the Circuit Using Logic Gates	05
	C	Solve using K-map F(A,B) = Σ(0,2,3) solve it by using K-Map and find digital logic circuit diagram.	05
	D	Minimize the given Boolean expression using a K-Map: $F(ABC) = ABC + \bar{A}BC + A\bar{B}C + AB\bar{C}$	05

- End of the Question Paper -